Vertical shell-and-tube condenser

Status: Completed with warnings or failed checks: review them

Preliminary design for coursework. Any mechanical sizing shown is a screening calculation from HEDH Part 4. Code compliance (ASME, TEMA, HEI, ISO) is not established by this report and needs qualified engineering review against the applicable edition.

Inputs

Input	Value	Unit
Configuration	in_tube_downflow	
Tube outside diameter	25.4	mm
Tube wall thickness	2.11	mm
Tube wall conductivity	18	W/m.K
Effective tube length available	4	m
Tube pitch	31.75	mm
Tube layout	90 deg square	
Tube material group for the TEMA span	A: steel and steel alloys	
Is the shell-side fluid fouling?	yes	
Bundle type	fixed tubesheet	
Bundle-to-shell diametral clearance L_bb	15	mm
Tube-to-baffle diametral clearance L_tb	0.8	mm
Sealing strip pairs per baffle	0	
Segmental baffle cut B_c	25	%
Baffle spacing / shell diameter	0.4	
Fluid name	n-pentane	
Saturation temperature	61.85	degC
Latent heat of condensation	337.3	kJ/kg
Liquid density	582.9	kg/m3
Vapour density	6.36	kg/m3
Liquid viscosity	0.000159	Pa.s
Vapour viscosity	7.6e-06	Pa.s
Liquid thermal conductivity	0.098	W/m.K
Liquid specific heat	2520	J/kg.K
Surface tension	0.0113	N/m
Condensing fluid mass flow	5	kg/s
Inlet quality	1	
Outlet quality	0	
Condensing pressure	2.27	bar
Vapour inlet temperature, if superheated	70	degC
Condensate outlet temperature, if subcooled	50	degC
Vapour specific heat	1960	J/kg.K
Vapour thermal conductivity	0.0193	W/m.K
Mixture condensation curve (optional, downflow)		
Non-condensable gas at the inlet	0	%
Fluid name	water	
Density	994.74	kg/m3
Specific heat	4178	J/kg.K
Viscosity	0.00075881	Pa.s
Thermal conductivity	0.62104	W/m.K
Coolant inlet temperature	25	degC
Coolant outlet temperature	40	degC
Fouling resistance, condensing side	0.0001	m2.K/W
Fouling resistance, coolant side	0.0002	m2.K/W
Allowable pressure drop, condensing side	20	kPa
Allowable pressure drop, coolant side	35	kPa
Design pressure, shell side	500	kPa
Design pressure, tube side	500	kPa
Allowable stress S	121	MPa
Weld joint efficiency E	0.85	
Corrosion allowance	3	mm
Tube metal	18Cr/8Ni stainless (S30400)	
Quality steps in the condensing zone	40	
Suction correction of the interfacial shear (HSB Eq. 16.76)	off	
Flooding tube-end factor F2 (reflux)	1	
Maximum iterations on the number of tubes	60	
Largest tube count tried	20000	

Results: thermal

Result	Value	Unit	Source
Heat duty	1.9157e+06	W	HSB Eq. 18.24
Coolant flow rate	30.568	kg/s	HSB Eq. 18.12
Coolant outlet temperature	40	degC	HSB Eq. 18.12
Area required, outside	78.664	m2	HSB Eq. 18.24
Dry-wall desuperheating zone area	0	m2	HSB Eq. 18.16
Condensing zone area	71.757	m2	HSB Eq. 18.19, Eq. 3.17
Subcooling zone area	6.9075	m2	HSB Eq. 18.23
Excess area	0.22205	%	
Overall coefficient, area-weighted mean	859.41	W/m2.K	HSB Eq. 18.21
Condensing coefficient, area-based mean	2446.5	W/m2.K	condensation_library.md
Condensing coefficient, first step	3274.5	W/m2.K	HSB Eq. 16.56
Condensing coefficient, last step	2212	W/m2.K	HSB Eq. 16.56
Coolant coefficient	4753.9	W/m2.K	HEDH 3.3.8
Product of shell-side correction factors	0.5579		HEDH 3.3.8
Vapour temperature where the wall becomes wet	70	degC	HSB Eq. 18.13
Area required, Boyko-Kruzhilin cross-check	80.421	m2	HSB Eq. 16.55
Area required, Chen sensitivity	85.924	m2	HSB Eq. 11.49
Area required, suction on	77.282	m2	HSB Eq. 16.76

Results: hydraulic

Result	Value	Unit	Source
Coolant velocity	0.9914	m/s	HEDH 3.3.5
Coolant Reynolds number	33011		HEDH 3.3.5
Coolant pressure drop, excluding nozzles	19394	Pa	HEDH 3.3.8 Eq. (7)
Coolant static rise over the tube length, not in the drop above	39034	Pa	Hydrostatics
Condensing-side mass flux	57.455	kg/m2.s	HSB Ex. 16.2
Vapour velocity at the inlet	9.0339	m/s	
Condensing-side drop against the allowable, recoveries not credited	752.55	Pa	HEDH 3.4.7
Two-phase friction	752.55	Pa	HSB Eq. 10.49
Acceleration term (negative: recovery)	-513.38	Pa	HSB Eq. 10.27
Gravity term (negative in downflow: gain)	-2455.5	Pa	HSB Eq. 10.26
Net pressure change in the tubes	-2216.3	Pa	HSB Eq. 10.27

Results: geometric

Result	Value	Unit	Source
Configuration	in_tube_downflow		HSB Sec. 17.3.1
Number of tubes	247		HEDH 3.4.9B
Tube passes	1		HEDH 3.4.3A
Tube count set by	required length		
Shell inside diameter (not rounded to a standard size)	0.6054	m	HEDH 3.3.5 Eqs. (16), (20), (21)
Bundle-to-shell clearance used	0.015	m	entered
Bundle diameter	0.5904	m	HEDH 3.3.5 Eq. (21)
Tube bore	0.02118	m	HEDH 3.3.5
Tube length available	4	m	
Tube length required	3.9911	m	HEDH 3.4.9B
Area available, outside	78.839	m2	HEDH 3.3.5 Eq. (30)
Tube-to-baffle clearance used	0.0008	m	entered
Shell-to-baffle clearance used	0.0055216	m	HEDH 3.3.5 Eq. (19)
Baffle spacing	0.24216	m	Lecture 9; HEDH 3.3.5
Number of baffles	15		HEDH 3.3.6 Eq. (14)
Next standard shell ID. The design is not re-run on it	0.635	m	HEDH 4.2.5 Table 2a

Results: mechanical

Result	Value	Unit	Source
Shell wall for pressure, without corrosion	0.0014905	m	HEDH 4.3.6 Eq. (8)
Shell wall to use	0.0095	m	HEDH 4.3.6 Table 2
Tube wall for internal pressure	4.3869e-05	m	HEDH 4.3.6 Eq. (8)
Tubesheet, rough estimate	0.06054	m	HEDH 3.3.5 Eq. (1)
Tube natural frequency, longest span	178.83	Hz	HEDH 4.6.2 Eq. (1)
Critical velocity, fluid elastic instability	6.0928	m/s	HEDH 4.6.4 Eq. (4)
Shell-side velocity for the vibration screen	1.064	m/s	HEDH 4.6.3
Vortex shedding frequency	17.384	Hz	HEDH 4.6.4 Eq. (2)
Turbulent buffeting frequency	10.777	Hz	HEDH 4.6.4 Eq. (3)

Engineering checks

Check	State	Detail	Source
Tube length covers the required length	PASS	L_req = 3.991 m <= L = 4.000 m	HEDH 3.4.9B
Smallest temperature difference in the condensing zone	PASS	21.9 K; Lecture 9: at least 3 to 5 K	Lecture 9
Largest temperature difference	PASS	45.0 K; Lecture 9 words this as 'at least 20 degC'	Lecture 9
Film thin against the tube radius (basis of the film model)	PASS	Largest shear-based film thickness / (D_i/2) = 0.241. HSB Ex. 16.2 assumes a thin film and no entrainment; this estimate overstates the thickness where gravity controls. The limit 0.5 is this module's judgement.	HSB Ex. 16.2; condensation_library.md
Shear stress nearly uniform across the film (tau_0 = tau_i), where shear controls	FAIL	Largest delta rho_L g / tau = 1.765 over the 18 of 40 steps with alpha_sh >= alpha_grav, HSB's check in Ex. 16.2. Over the others gravity carries the film and Eq. 16.56 relies on its gravity term.	HSB Ex. 16.2
Film regimes met along the tube	NOT ASSESSED	laminar wavy, turbulent; Re_f from 96 to 7558. Gravity branch by regime, HEDH 2.6.2. Pr_L = 4.09; Pr is capped at 10 in the turbulent branch.	HEDH 2.6.2
Boyko-Kruzhilin cross-check on the area	PASS	A_BK = 80.42 m2 against A = 78.66 m2, +2.2%; Chen +9.2%; other suction setting -1.8%. The sources give no agreement criterion; 25% is this module's flag for a closer look.	HSB Sec. 16.3.2.2
Premoli slip bracket	PASS	0 of 40 steps have a negative bracket, where the library returns S = 1. HSB is silent. Affects only the acceleration and gravity terms.	HSB Eqs. 10.59-10.64; condensation_library.md
Dittus-Boelter range, vapour in the dry zone	PASS	Re_v = 160119 > 1e4; 0.7 <= Pr_v = 0.77 <= 160; L/D = 189 > 60. The entry length of the zone itself is not checked.	singlephase.py; ME6220 Tutorial 1
Baffle cut in the range of the J_c fit	PASS	B_c = 25%; 15 to 45%	HEDH 3.3.6I
Baffle cut against spacing, HEDH Fig. 3.3.5(4), SBC band	PASS	B_c = 25% at L_bc/D_s = 0.40; band 21.2 to 25.9%, centre line 24.0%. Chart read +-1% B_c, allowed for. SBC is the no-phase-change band, the coolant here.	HEDH 3.3.5 Fig. 4, digitised
Baffle leakage correction not too low	PASS	J_l = 0.635 >= 0.6; preferably 0.7 to 0.9	HEDH 3.3.6J
Product of correction factors	PASS	J_c J_l J_b J_r = 0.558 >= 0.4; preferably >= 0.5	HEDH 3.3.8A
Coolant in turbulent shell-side flow	PASS	Re_s = 33011. Below 100 HSB doubts the method's validity and HEDH applies J_r.	HSB Sec. 6.3.2; HEDH 3.3.6L
Kern is valid only at 25% cut	PASS	B_c = 25%. Kern is the cross-check only.	HSB Sec. 6.3.1
Number of baffles	PASS	N_b = 15 >= 5. HEDH 3.3.6M: fewer spacings is questionable.	HEDH 3.3.6M
Baffle spacing at least D_s/5 and at least 50.8 mm	PASS	L_bc = 242.2 mm; limit 121.1 mm	Lecture 9; HSB Sec. 6.2.6; HEDH 3.3.5 item 12
Baffle spacing at most D_s	PASS	L_bc = 242.2 mm <= 605.4 mm	HEDH 3.3.5 item 12
Kern cross-check on the coolant coefficient	NOT ASSESSED	h_Kern = 4799 against Bell-Delaware 4754 W/m2.K, ratio 1.01. HSB's Ex. 6.1 and 6.2 give 978 and 769 on one exchanger, so such a gap is expected.	HSB Exs. 6.1, 6.2
Pitch at least 1.25 D_o (TEMA, via Lecture 9)	PASS	L_tp/D_o = 1.250 >= 1.25	Lecture 9; HSB Sec. 6.2.5
Pitch ratio within the shell-side method	PASS	L_tp/D_o = 1.250; 1.25 to 1.5	HEDH 3.3.5 item 6
Lane between tubes with a fouling shell-side fluid	PASS	Gap = 6.35 mm; at least 6.35 mm if fouling (stated as yes).	Lecture 9; HSB Sec. 6.2.5
Web between tube holes	PASS	Gap = 6.35 mm >= 3.2 mm	HSB Sec. 6.2.5
Tube diameter for cleaning	PASS	D_o = 25.40 mm; HSB and HEDH: about 20 mm minimum where cleaning is needed. 19.05 mm is the usual tube, so 19 mm is the limit used.	HSB Sec. 6.2.3; HEDH 3.3.5 item 2
Tube length in the usual range	PASS	L = 4 m; Lecture 9: 3.6 to 11 m	Lecture 9
Length to shell diameter ratio	PASS	L/D_s = 6.6; HSB: 5 to 10 for best all-round performance	HSB Sec. 6.2.4
Shell to tube diameter ratio	PASS	D_s/D_o = 23.8 >= 15; the shell-side and count methods lose accuracy below this	HEDH 3.3.5 item 2
Tubes per pass	PASS	N = 247 >= 8	HEDH 3.3.5 item 16
Tube wall thickness against the Lecture 9 table	PASS	t = 2.11 mm, 18Cr/8Ni stainless (S30400). Lecture 9 and HEDH 4.2.5-1, carbon and low alloy or stainless and copper alloys: OD 15.88: 1.65 or 1.24; 19.05: 2.11 or 1.65; 25.40: 2.77 or 2.11; 31.75: 3.40 or 2.77 mm. Lecture 9 misprints the third OD as 24.40 and 25.04; HEDH settles it as 25.40 mm (1 in). HEDH calls these typical; the pressure check is separate. At least 2.11 mm here.	Lecture 9; HEDH 4.2.5-1
Unsupported span in the window, 2 L_bc, within the TEMA maximum (as quoted in HEDH Part 4)	PASS	2 L_bc = 484 mm <= L_b,max = 1853 mm. HEDH's Eqs. (10) run up to 2.3% below the TEMA table quoted in HEDH 4.2.5-7 (1308, 1523, 1853, 2183 mm against 1321, 1524, 1880, 2235 mm), so they err on the safe side. HEDH 4.6.2 and HSB: the current trend is to limit spans to 80% of the TEMA value, here 26%.	HEDH 3.3.5 Eqs. (10); HEDH 4.2.5-7; HSB Table 6.1; Lecture 9

Check	State	Detail	Source
Condensing pressure drop within the allowable	PASS	dP = 0.75 kPa <= 20.00 kPa	HSB Sec. 18.4
Coolant pressure drop within the allowable	PASS	dP = 19.39 kPa <= 35.00 kPa, nozzles excluded	HSB Sec. 18.4
Coolant velocity, shell side	PASS	u = 0.99 m/s; Lecture 9: 0.3 to 1 m/s for single phase. Based on S_m.	Lecture 9
Flow distribution among parallel tubes	NOT ASSESSED	Inlet velocity head 260 Pa against tube friction 753 Pa. HSB Sec. 18.2.1: tube drop may not be monotonic in flow, so maldistribution and instability are possible. HEDH 3.4.3A: compare the nozzle velocity head with the tube drop. No numeric criterion is given.	HSB Sec. 18.2.1; HEDH 3.4.3A
Pressure-drop feedback on the condensing temperature	NOT ASSESSED	dP/P = 0.33%. dT_sat ~ T (1/rho_v - 1/rho_L) dP / h_fg = 0.12 K (Clapeyron, a thermodynamic identity, not from the sources). HEDH 3.4.9B step 7: correct T_sat 'if the pressure drop is large enough', with no value. Not iterated. Smallest dT in the zone: 21.9 K.	HEDH 3.4.9B; Lecture 24
Coolant nozzle size	NOT ASSESSED	Minimum inside diameter 161 mm at rho v^2 = 2250 kg/m.s2 with no impingement plate (HEDH Eqs. 3.3.5(11a), (15), TEMA as quoted; HEDH 4.2.5-7 prints 2230). Nozzle pressure drop and selection are not assessed; HEDH says vibration may require larger.	HEDH 3.3.5 item 14
Venting of non-condensables	NOT ASSESSED	In-tube flow has a fixed path, so venting is simple and positive (HEDH 3.4.3A, 3.4.5B). Drill vents in the upper tubesheet for the shell-side coolant, discharging continuously to an open drain, and fit a funnel or baffle in the lower head to limit re-entrainment into the vent gas (HEDH 3.4.3A). HSB Sec. 17.1: arrange purging for start-up. No vent size is calculated: HEDH 3.4.5B calls the literature sparse with very few rules, and no source gives a vent rate.	HEDH 3.4.3, 3.4.5B; HSB Secs. 17.1, 17.3.1, 18.5; Lecture 23
Fogging	NOT ASSESSED	Relevant only with non-condensables or mixtures (HEDH 3.4.5C, 2.6.7). No source gives a design number, so it is not calculated. HEDH's measures: keep the temperature difference small, and fit demisters if fog is likely.	HEDH 3.4.5C, 2.6.7F
Tube length within the common range of HEDH 4.2.5	PASS	L = 4 m; HEDH 4.2.5-2: most commonly 3.6 to 15 m for this bundle type, and up to 22 m has been made. Neither book lists standard lengths.	HEDH 4.2.5-2
Standard shell holds the tubes, HEDH 4.2.5 Table 2a	NOT ASSESSED	Next shell 635 mm against D_s = 605 mm. No count for this layout. Table 2a covers fixed tubesheets with one of 15.88 x 19.84, 19.05 x 23.81, 19.05 x 25.4, 25.4 x 31.75, 31.75 x 39.69 mm on a triangular pitch. Tables 2b and 2c, with the square pitches, are not coded.	HEDH 4.2.5 Table 2a
Tube wall against internal pressure	PASS	t = 2.11 mm >= 0.044 mm, S = 121 MPa (HEDH 4.3.6, SA-515-70 carbon steel plate at 100 degC). External pressure on the tube needs an ASME chart factor that HEDH prints only for its example (Eq. 10); HEDH 4.2.5-1 gives as a guide that a 19 mm x 2.1 mm steel tube withstands 16 MN/m2 internal and 7 MN/m2 external. Preliminary sizing from HEDH Part 4, which quotes ASME Section VIII and TEMA. It does not establish code compliance.	HEDH 4.3.6 Sec. D; HEDH 4.2.5-1
Tubesheet shear does not control	NOT ASSESSED	No criterion for the rough estimate. A fixed tubesheet carries thermal loads as well as pressure (HEDH 4.3.2I), and neither is in 0.1 D_s.	HEDH 4.3.2I
Pressure boundary: shell, tube and tubesheet	NOT ASSESSED	Sized, no verdict: shell 1.49 mm + 3 mm corrosion, TEMA class R minimum 9.5 mm, use 9.50 mm; tubesheet about 61 mm by the rough 0.1 D_s. Preliminary sizing from HEDH Part 4, which quotes ASME Section VIII and TEMA. It does not establish code compliance. ASME and TEMA are as quoted in HEDH, 1977/1986 and 1978 editions. Expansion joint, flanges and nozzle reinforcement are not sized.	HEDH 4.2.6, 4.3.2, 4.3.6
Tube vibration: fluid elastic instability	PASS	u_c = 1.064 m/s < u_crit = 6.093 m/s, with the lowest beta band. Central baffle spaces only; HEDH asks for inlet, central and outlet regions, and the nozzles are not sized here. HEDH: vibration and damage are probable if u_c exceeds u_crit.	HEDH 4.6.5A step 2
Tube vibration: vortex shedding	PASS	f_vs = 17.4 Hz <= 0.8 f_n = 143.1 Hz. HEDH: above this vibration is possible, not necessarily damage. Vortex shedding may not exist deep in a tightly spaced bundle.	HEDH 4.6.5A step 3
Tube vibration: turbulent buffeting	PASS	f_tb = 10.8 Hz <= 0.8 f_n = 143.1 Hz. HEDH: Eq. (3) comes from gas data and may not apply to a liquid.	HEDH 4.6.5A step 4

Calculation steps (formula = numbers = result)

Step	Formula	Numbers	Result	Source
Latent duty	Q_lat = M_v h_LG (x_in - x_out)	5 (3.373e+05) (1 - 0)	Q_lat = 1.6865e+06 W	HSB Ex. 19.1
Total superheat duty, inlet to saturation	Q_sup = M_v c_pv (T_h,in - T_sat)	5 (1960) (70 - 61.85)	Q_sup = 79870 W	HSB Eqs. 18.15, 18.17; Lecture 24
Subcooling duty	Q_sub = M_v c_pL (T_sat - T_h,out)	5 (2520) (61.85 - 50)	Q_sub = 1.4931e+05 W	HSB Eq. 18.22
Total duty	Q = Q_lat + Q_sup + Q_sub	1.6865e+06 + 79870 + 1.4931e+05	Q = 1.9157e+06 W	HSB Eq. 18.24
Coolant flow: the outlet temperature is given, so the heat balance gives the flow	M_c = Q / [c_pc (T_c,out - T_c,in)]	1.9157e+06 / [4178 (40 - 25)]	M_c = 30.568 kg/s	HSB Eq. 18.12
Tube bore	D_i = D_o - 2 t	0.0254 - 2 (0.00211)	D_i = 0.02118 m	HEDH 3.3.5 item 4
Number of tubes, set by: required length. See the iteration table	N = smallest count meeting L_req <= L and the entered limits	iteration table below	N = 247	HEDH 3.4.9B; HSB Sec. 18.3.3
Diameter through the centres of the outer tubes, inverting the tube count estimate. One tube pass, so no pass correction	N = 0.78 D_ctl^2 / (C1 L_tp^2), so D_ctl = L_tp (N C1 / 0.78)^(1/2)	0.03175 (247 1 / 0.78)^0.5	D_ctl = 0.565 m	HEDH 3.3.5 Eq. (16)
Bundle diameter	D_otl = D_ctl + D_o	0.565 + 0.0254	D_otl = 0.5904 m	HEDH 3.3.5 Eq. (21)
Shell inside diameter	D_s = D_otl + L_bb	0.5904 + 0.015	D_s = 0.6054 m	HEDH 3.3.5 Eq. (20a)
Available area, outside of the tubes	A = pi D_o L N	pi (0.0254) (4) (247)	A = 78.839 m2	HEDH 3.3.5 Eq. (30)
Wall resistance, referred to the outside	R_w = D_o ln(D_o/D_i) / (2 k)	0.0254 ln(0.0254/0.02118) / (2 18)	R_w = 0.00012819 m2.K/W	HSB Eq. 18.11
Condensing-side mass flux in one tube	G = M_v / (N pi D_i^2 / 4)	5 / (247 pi 0.02118^2 / 4)	G = 57.455 kg/m2.s	HSB Ex. 16.2
Coolant side: Bell-Delaware, not Kern. HEDH gives it in equations and HSB and Lecture 12 call it the widely accepted, more accurate method; Kern is restricted to a 25% cut and has no equation for f. Kern's coefficient is the cross-check below	L_bc = absolute spacing, or max(ratio D_s, 50.8 mm), limited to L_b,max / 2	ratio 0.4, D_s = 0.6054 m	L_bc = 0.24216 m	Lecture 9; HEDH 3.3.5 item 12
Cross-flow area at the shell centreline	S_m = L_bc [L_bb + (D_ctl / L_tp,eff)(L_tp - D_o)]	0.24216 [0.015 + (0.565 / 0.03175)(0.03175 - 0.0254)]	S_m = 0.030996 m2	HEDH 3.3.5 Eq. (22)
Shell-to-baffle clearance left blank: the TEMA curve plus 1.5 mm for out-of-roundness, as HEDH captions Fig. 13	L_sb = 3.1 mm + 0.004 D_s	3.1e-3 + 0.004 (0.6054)	L_sb = 0.0055216 m	HEDH 3.3.5 Eq. (19)
Shell-side mass velocity	m_s = M_c / S_m	30.568 / 0.030996	m_s = 986.18 kg/m2.s	HEDH 3.3.5 Eq. (24)
Shell-side Reynolds number	Re_s = D_o m_s / mu	0.0254 (986.18) / 0.00075881	Re_s = 33011	HEDH 3.3.5 Eq. (25)
Shell-side Prandtl number	Pr_s = c_p mu / k	4178 (0.00075881) / 0.62104	Pr_s = 5.1048	HEDH 3.3.5 Eq. (26)
Baffle cut angle at the shell	theta_ds = 2 acos(1 - 2 B_c/100)	2 acos(1 - 2 (25)/100)	theta_ds = 120 deg	HEDH 3.3.6 Eq. (1)
Baffle cut angle at D_ctl	theta_ctl = 2 acos[(D_s/D_ctl)(1 - 2 B_c/100)]	2 acos[(0.6054/0.565)(1 - 2 (25)/100)]	theta_ctl = 115.21 deg	HEDH 3.3.6 Eq. (2)
Fraction of tubes in one window	F_w = theta_ctl/360 - sin(theta_ctl)/(2 pi)	115.21/360 - sin(115.21 deg)/(2 pi)	F_w = 0.17603	HEDH 3.3.6 Eq. (5)
Fraction of tubes in cross flow	F_c = 1 - 2 F_w	1 - 2 (0.17603)	F_c = 0.64794	HEDH 3.3.6 Eq. (6)
Gross window area	S_wg = (pi/4) D_s^2 [theta_ds/360 - sin(theta_ds)/(2 pi)]	(pi/4) 0.6054^2 [120/360 - sin(120 deg)/(2 pi)]	S_wg = 0.056275 m2	HEDH 3.3.6 Eq. (4)
Window area taken by tubes	S_wt = N F_w (pi/4) D_o^2	247 (0.17603) (pi/4) 0.0254^2	S_wt = 0.022032 m2	HEDH 3.3.6 Eq. (7)
Net window area	S_w = S_wg - S_wt	0.056275 - 0.022032	S_w = 0.034244 m2	HEDH 3.3.6 Eq. (9)
Tube rows crossed between baffle tips	N_tcc = (D_s/L_pp)(1 - 2 B_c/100)	(0.6054/0.03175)(1 - 2 (25)/100)	N_tcc = 9.5338	HEDH 3.3.6 Eq. (11)
Effective tube rows crossed in one window	N_tcw = (0.8/L_pp)[D_s B_c/100 - (D_s - D_ctl)/2]	(0.8/0.03175)[0.6054 (25)/100 - (0.6054 - 0.565)/2]	N_tcw = 3.3045	HEDH 3.3.6 Eq. (13)
Number of baffles, rounded down	N_b = L_ti/L_bc - 1	4/0.24216 - 1	N_b = 15	HEDH 3.3.6 Eq. (14)

Step	Formula	Numbers	Result	Source
Bypass area per baffle space. No tube-pass lane	S_b = L_bc (D_s - D_otl)	0.24216 (0.6054 - 0.5904)	S_b = 0.0036324 m2	HEDH 3.3.6 Eq. (15)
Bypass fraction	F_sbp = S_b/S_m	0.0036324/0.030996	F_sbp = 0.11719	HEDH 3.3.6 Eq. (16)
Shell-to-baffle leakage area	S_sb = pi D_s (L_sb/2)(360 - theta_ds)/360	pi (0.6054) (0.0055216/2) (360 - 120)/360	S_sb = 0.0035005 m2	HEDH 3.3.6 Eq. (17)
Tube-to-baffle leakage area	S_tb = (pi/4)[(D_o + L_tb)^2 - D_o^2] N (1 - F_w)	(pi/4)[(0.0254 + 0.0008)^2 - 0.0254^2] (247) (1 - 0.17603)	S_tb = 0.0065984 m2	HEDH 3.3.6 Eq. (18)
Leakage parameters	r_lm = (S_sb + S_tb)/S_m; r_s = S_sb/(S_sb + S_tb)	(0.0035005 + 0.0065984)/0.030996; r_s = 0.34662	r_lm = 0.32581	HEDH 3.3.6 Eqs. (21), (22)
Ideal tube bank j factor, Table 1 row for this Re and layout	j_i = a1 (1.33/(L_tp/D_o))^a Re^a2, a = a3/(1 + 0.14 Re^a4)	Re = 33011, L_tp/D_o = 1.25, 90 deg square	j_i = 0.0061313	HEDH 3.3.7 Eq. (1), Table 1
Viscosity correction, liquid. Ratio 1 if no wall viscosity is entered	phi^r = (mu/mu_w)^0.14	(0.00075881/0.00075881)^0.14	phi^r = 1	HEDH 3.3.7 Eq. (5)
Ideal tube bank coefficient	alpha_i = j_i c_p m_s Pr^(-2/3) phi^r	0.0061313 (4178) (986.18) (5.1048)^(-2/3) (1)	alpha_i = 8521 W/m2.K	HEDH 3.3.7 Eq. (4)
Baffle window correction	J_c = 0.55 + 0.72 F_c	0.55 + 0.72 (0.64794)	J_c = 1.0165	HEDH 3.3.6 Eq. (20); HSB Eq. 6.14
Baffle leakage correction	J_l = 0.44 (1 - r_s) + [1 - 0.44 (1 - r_s)] exp(-2.2 r_lm)	0.44 (1 - 0.34662) + [1 - 0.44 (1 - 0.34662)] exp(-2.2 (0.32581))	J_l = 0.63542	HEDH 3.3.6 Eq. (23a)
Bundle bypass correction	J_b = exp[-C_bh F_sbp (1 - (2 r_ss)^(1/3))], r_ss = N_ss/N_tcc	exp[-1.25 (0.11719) (1 - (2 (0))^(1/3))]	J_b = 0.86374	HEDH 3.3.6 Eqs. (24), (25)
Coolant coefficient. Equal end spacings, so J_s = 1	h_c = alpha_i J_c J_l J_b J_r	8521 (1.0165) (0.63542) (0.86374) (1)	h_c = 4753.9 W/m2.K	HEDH 3.3.8 Eq. (1)
Cross-check only: Kern, restricted to a 25% cut. Not used in the sizing	h = 0.36 (k/D_e) (D_e m_s/mu)^0.55 Pr^(1/3) (mu/mu_w)^0.14, m_s = M_c P_T/(D_s C L_bc)	0.36 (0.62104/0.025132) (34529)^0.55 (5.1048)^(1/3)	h_Kern = 4799.5 W/m2.K	HSB Eqs. 6.1, 6.3, 6.5, 6.6; Lecture 11
Ideal tube bank friction factor	f_i = b1 (1.33/(L_tp/D_o))^b Re^b2, b = b3/(1 + 0.14 Re^b4)	Re = 33011, L_tp/D_o = 1.25	f_i = 0.087952	HEDH 3.3.7 Eq. (2), Table 1
Ideal drop in one cross-flow section	dP_bi = 2 f_i N_tcc m_s^2 / rho / phi^r	2 (0.087952) (9.5338) (986.18)^2 / 994.74 / 1	dP_bi = 1639.6 Pa	HEDH 3.3.7 Eq. (10)
Leakage correction for pressure drop	R_l = exp[-1.33 (1 + r_s) r_lm^p], p = -0.15 (1 + r_s) + 0.8	exp[-1.33 (1 + 0.34662) (0.32581)^p], r_s = 0.34662	R_l = 0.40016	HEDH 3.3.6 Eq. (23b)
Bypass correction for pressure drop	R_b = exp[-C_bp F_sbp (1 - (2 r_ss)^(1/3))]	exp[-3.7 (0.11719) (1 - (2 (0))^(1/3))]	R_b = 0.64817	HEDH 3.3.6 Eq. (26)
Cross-flow drop between baffle tips	dP_c = dP_bi (N_b - 1) R_b R_l	1639.6 (15 - 1) (0.64817) (0.40016)	dP_c = 5953.8 Pa	HEDH 3.3.8 Eq. (2)
Window mass velocity	m_w = M_c / (S_m S_w)^(1/2)	30.568 / (0.030996 0.034244)^0.5	m_w = 938.25 kg/m2.s	HEDH 3.3.8 Eq. (3)
Window drop. Re >= 100	dP_w = N_b (2 + 0.6 N_tcw) m_w^2/(2 rho) R_l	15 (2 + 0.6 (3.3045)) (938.25)^2 / (2 994.74) (0.40016)	dP_w = 10578 Pa	HEDH 3.3.8 Eq. (4)
End-zone drop. Equal spacings, so R_s = 2	dP_e = dP_bi (1 + N_tcw/N_tcc) R_b R_s	1639.6 (1 + 3.3045/9.5338) (0.64817) (2)	dP_e = 2862.3 Pa	HEDH 3.3.8 Eq. (6)
Coolant pressure drop, excluding nozzles	dP_s = dP_c + dP_w + dP_e	5953.8 + 10578 + 2862.3	dP_coolant = 19394 Pa	HEDH 3.3.8 Eq. (7)
Zone 1: vapour Reynolds number	Re_v = G D_i / mu_v	57.455 (0.02118) / 7.6e-06	Re_v = 1.6012e+05	HSB Sec. 18.3.2
Zone 1: single-phase vapour coefficient. Being cooled, so n = 0.3	alpha_v = 0.023 (k_v/D_i) Re_v^0.8 Pr_v^0.3	0.023 (0.0193/0.02118) (1.6012e+05)^0.8 (0.77181)^0.3	alpha_v = 282.6 W/m2.K	singlephase.py; HSB Sec. 18.4.1
Zone 1: overall coefficient, outside area	1/U_s = 1/h_c + R_f,c + R_w + (D_o/D_i) R_f,v + (D_o/D_i)/alpha_v	h_c = 4753.9, alpha_v = 282.6, D_o/D_i = 1.1992	U_s = 203.99 W/m2.K	HSB Eq. 18.11, vapour side inside
Zone 1: ratio in Eq. 18.13, with U and alpha on the same (inside) area	U/alpha = U_s (D_o/D_i) / alpha_v	203.99 (1.1992) / 282.6	U/alpha = 0.86568	HSB Eq. 18.10
Vapour temperature at which the wall reaches T_sat and condensation begins. It is at or above the inlet temperature, so the wall is wet from the inlet and there is no dry zone	T_h* = [T_sat - (U/alpha)(T_c,out - T_h,in R)] / [1 - (U/alpha)(1 - R)], R = M_v c_pv/(M_c c_pc)	[61.85 - 0.86568 (40 - 70 (0.076735))] / [1 - 0.86568 (1 - 0.076735)]	T_h* = 158.77 degC	HSB Eq. 18.13; Lecture 24
Zone 2 load: latent plus wet-wall desuperheating	Q_con = M_v [c_pv (T_h* - T_sat) + h_LG]	5 [1960 (70 - 61.85) + 3.373e+05]	Q_con = 1.7664e+06 W	HSB Eq. 18.17
Coolant temperature at the zone 3 boundary	T_c' = T_c,in + (T_sat - T_h,out) M_v c_pL / (M_c c_pc)	25 + (61.85 - 50) 12600 / 1.2771e+05	T_c' = 26.169 degC	HSB Eq. 18.20
Film flow per unit periphery, first step, x = 0.9875. All liquid in the film	Gamma = G (1 - x) D_i / 4	57.455 (1 - 0.9875) (0.02118) / 4	Gamma = 0.0038028 kg/m.s	HSB Ex. 16.2 step 2(ii)

Step	Formula	Numbers	Result	Source
Film Reynolds number, first step, x = 0.9875	Re_f = 4 Gamma / eta_L	4 (0.0038028) / 0.000159	Re_f = 95.669	HSB Eq. 16.19
Gravity term, first step, x = 0.9875: laminar wavy	alpha_grav = (lambda_L/[eta^2/(rho_L (rho_L - rho_G) g)]^(1/3)) 0.756 Re_f^-0.22	lambda_L = 0.098, [eta^2/(rho_L (rho_L - rho_G) g)]^(1/3) = 1.972e-05, Re_f = 95.669, Pr_L = 4.0886	alpha_grav = 1377.5 W/m2.K	HSB Sec. 16.3.1; HEDH 2.6.2 Eqs. 14, 17, 20
Friedel multiplier, first step, x = 0.9875	phi_LO^2 = E + 3.24 F H / (Fr^0.045 We^0.035)	41.789 + 3.24 (0.34594) (33.094) / ((383.13)^0.045 (960.84)^0.035)	phi_LO^2 = 64.107	HSB Eqs. 10.49 to 10.55
Frictional gradient, first step, x = 0.9875	-dp_F/dz = phi_LO^2 (dp_F/dz)_LO	64.107 (4.5168)	-dp_F/dz = 289.56 Pa/m	HSB Eq. 10.49
Interfacial shear, first step, x = 0.9875	tau = (-dp_F/dz) D_i / 4	289.56 (0.02118) / 4	tau = 1.5332 N/m2	HSB Eq. 16.53
Dimensionless film thickness and temperature, first step, x = 0.9875	delta+ = 0.6323 Re_f^0.5286; T+ = 5 [Pr + ln(1 + Pr (delta+/5 - 1))]	delta+ = 7.0463, Pr_L = 4.0886	T+ = 25.359	HSB Eqs. 16.46 to 16.51
Shear term, Kosky-Staub, first step, x = 0.9875	alpha_sh = c_pL (rho_L tau)^(1/2) / T+	2520 (582.9 1.5332)^0.5 / 25.359	alpha_sh = 2970.7 W/m2.K	HSB Eq. 16.52
Combined coefficient, first step, x = 0.9875	alpha = (alpha_sh^2 + alpha_grav^2)^(1/2)	(2970.7^2 + 1377.5^2)^0.5	alpha = 3274.5 W/m2.K	HSB Eq. 16.56
Overall coefficient, first step, x = 0.9875, outside area	1/U = 1/h_o + R_f,o + R_w + (D_o/D_i) R_f,i + (D_o/D_i)/h_i	h_c = 4753.9, alpha = 3274.5, R_f,cond = 0.0001, R_f,cool = 0.0002, D_o/D_i = 1.1992	U = 975.89 W/m2.K	HSB Eq. 18.21
Local temperature difference, first step, x = 0.9875	dT = log mean of (T_sat - T_c) at the step ends	61.85 - 40, 61.85 - 39.654	dT = 22.022 K	HSB Eq. 3.17; Secs. 18.2.1, 18.3.1
Step area, first step, x = 0.9875	dA = dQ / (U dT)	44159 / (975.89 22.022)	dA = 2.0547 m2	HSB Eq. 3.17; Secs. 18.2.1, 18.3.1
Film flow per unit periphery, middle step, x = 0.4875. All liquid in the film	Gamma = G (1 - x) D_i / 4	57.455 (1 - 0.4875) (0.02118) / 4	Gamma = 0.15592 kg/m.s	HSB Ex. 16.2 step 2(ii)
Film Reynolds number, middle step, x = 0.4875	Re_f = 4 Gamma / eta_L	4 (0.15592) / 0.000159	Re_f = 3922.4	HSB Eq. 16.19
Gravity term, middle step, x = 0.4875: turbulent	alpha_grav = (lambda_L/[eta^2/(rho_L (rho_L - rho_G) g)]^(1/3)) 0.023 Re_f^0.25 Pr_L^0.5	lambda_L = 0.098, [eta^2/(rho_L (rho_L - rho_G) g)]^(1/3) = 1.972e-05, Re_f = 3922.4, Pr_L = 4.0886	alpha_grav = 1829.1 W/m2.K	HSB Sec. 16.3.1; HEDH 2.6.2 Eqs. 14, 17, 20
Friedel multiplier, middle step, x = 0.4875	phi_LO^2 = E + 3.24 F H / (Fr^0.045 We^0.035)	10.447 + 3.24 (0.48635) (33.094) / ((95.501)^0.045 (479.71)^0.035)	phi_LO^2 = 44.669	HSB Eqs. 10.49 to 10.55
Frictional gradient, middle step, x = 0.4875	-dp_F/dz = phi_LO^2 (dp_F/dz)_LO	44.669 (4.5168)	-dp_F/dz = 201.76 Pa/m	HSB Eq. 10.49
Interfacial shear, middle step, x = 0.4875	tau = (-dp_F/dz) D_i / 4	201.76 (0.02118) / 4	tau = 1.0683 N/m2	HSB Eq. 16.53
Dimensionless film thickness and temperature, middle step, x = 0.4875	delta+ = 0.0504 Re_f^0.875; T+ = 5 [Pr + ln(1 + 5 Pr) + 0.5 ln(delta+/30)]	delta+ = 70.273, Pr_L = 4.0886	T+ = 37.898	HSB Eqs. 16.46 to 16.51
Shear term, Kosky-Staub, middle step, x = 0.4875	alpha_sh = c_pL (rho_L tau)^(1/2) / T+	2520 (582.9 1.0683)^0.5 / 37.898	alpha_sh = 1659.3 W/m2.K	HSB Eq. 16.52
Combined coefficient, middle step, x = 0.4875	alpha = (alpha_sh^2 + alpha_grav^2)^(1/2)	(1659.3^2 + 1829.1^2)^0.5	alpha = 2469.6 W/m2.K	HSB Eq. 16.56
Overall coefficient, middle step, x = 0.4875, outside area	1/U = 1/h_o + R_f,o + R_w + (D_o/D_i) R_f,i + (D_o/D_i)/h_i	h_c = 4753.9, alpha = 2469.6, R_f,cond = 0.0001, R_f,cool = 0.0002, D_o/D_i = 1.1992	U = 874.07 W/m2.K	HSB Eq. 18.21
Local temperature difference, middle step, x = 0.4875	dT = log mean of (T_sat - T_c) at the step ends	61.85 - 33.085, 61.85 - 32.739	dT = 28.938 K	HSB Eq. 3.17; Secs. 18.2.1, 18.3.1
Step area, middle step, x = 0.4875	dA = dQ / (U dT)	44159 / (874.07 28.938)	dA = 1.7459 m2	HSB Eq. 3.17; Secs. 18.2.1, 18.3.1
Film flow per unit periphery, last step, x = 0.0125. All liquid in the film	Gamma = G (1 - x) D_i / 4	57.455 (1 - 0.0125) (0.02118) / 4	Gamma = 0.30042 kg/m.s	HSB Ex. 16.2 step 2(ii)
Film Reynolds number, last step, x = 0.0125	Re_f = 4 Gamma / eta_L	4 (0.30042) / 0.000159	Re_f = 7557.8	HSB Eq. 16.19
Gravity term, last step, x = 0.0125: turbulent	alpha_grav = (lambda_L/[eta^2/(rho_L (rho_L - rho_G) g)]^(1/3)) 0.023 Re_f^0.25 Pr_L^0.5	lambda_L = 0.098, [eta^2/(rho_L (rho_L - rho_G) g)]^(1/3) = 1.972e-05, Re_f = 7557.8, Pr_L = 4.0886	alpha_grav = 2155 W/m2.K	HSB Sec. 16.3.1; HEDH 2.6.2 Eqs. 14, 17, 20
Friedel multiplier, last step, x = 0.0125	phi_LO^2 = E + 3.24 F H / (Fr^0.045 We^0.035)	0.98185 + 3.24 (0.03268) (33.094) / ((0.21277)^0.045 (22.643)^0.035)	phi_LO^2 = 4.35	HSB Eqs. 10.49 to 10.55

Step	Formula	Numbers	Result	Source
Frictional gradient, last step, x = 0.0125	-dp_F/dz = phi_LO^2 (dp_F/dz)_LO	4.35 (4.5168)	-dp_F/dz = 19.648 Pa/m	HSB Eq. 10.49
Interfacial shear, last step, x = 0.0125	tau = (-dp_F/dz) D_i / 4	19.648 (0.02118) / 4	tau = 0.10404 N/m2	HSB Eq. 16.53
Dimensionless film thickness and temperature, last step, x = 0.0125	delta+ = 0.0504 Re_f^0.875; T+ = 5 [Pr + ln(1 + 5 Pr) + 0.5 ln(delta+/30)]	delta+ = 124.75, Pr_L = 4.0886	T+ = 39.333	HSB Eqs. 16.46 to 16.51
Shear term, Kosky-Staub, last step, x = 0.0125	alpha_sh = c_pL (rho_L tau)^(1/2) / T+	2520 (582.9 0.10404)^0.5 / 39.333	alpha_sh = 498.93 W/m2.K	HSB Eq. 16.52
Combined coefficient, last step, x = 0.0125	alpha = (alpha_sh^2 + alpha_grav^2)^(1/2)	(498.93^2 + 2155^2)^0.5	alpha = 2212 W/m2.K	HSB Eq. 16.56
Overall coefficient, last step, x = 0.0125, outside area	1/U = 1/h_o + R_f,o + R_w + (D_o/D_i) R_f,i + (D_o/D_i)/h_i	h_c = 4753.9, alpha = 2212, R_f,cond = 0.0001, R_f,cool = 0.0002, D_o/D_i = 1.1992	U = 832.89 W/m2.K	HSB Eq. 18.21
Local temperature difference, last step, x = 0.0125	dT = log mean of (T_sat - T_c) at the step ends	61.85 - 26.515, 61.85 - 26.169	dT = 35.508 K	HSB Eq. 3.17; Secs. 18.2.1, 18.3.1
Step area, last step, x = 0.0125	dA = dQ / (U dT)	44159 / (832.89 35.508)	dA = 1.4932 m2	HSB Eq. 3.17; Secs. 18.2.1, 18.3.1
Zone 2 area, sum over the steps. T_c at the zone ends is linear in Q	A_con = sum dA	sum of 40 steps, table below	A_con = 71.757 m2	HSB Eq. 3.17; Secs. 18.2.1, 18.3.1
Zone 3: film flow per unit periphery	Gamma = M_v / (N pi D)	5 / (247 pi 0.02118)	Gamma = 0.30423 kg/m.s	HEDH 3.4.6G
Zone 3: film regime. Re_u = 2460 Pr^-0.65 = 984.95	Re_L = 4 Gamma / eta_L	4 (0.30423) / 0.000159	Re_L = 7653.5	HEDH 3.4.6G
Zone 3: subcooled film coefficient, turbulent, Eq. (33)	alpha (eta^2/(g rho^2))^(1/3) / lambda = 5.7e-3 Re^0.4 Pr^0.34	lambda_L = 0.098, Re_L = 7653.5, Pr_L = 4.0886, L_sub = 0.35046 m	alpha_sc = 1641.6 W/m2.K	HEDH 3.4.6 Eqs. (31)-(33); Lecture 24
Zone 3: overall coefficient, outside area	1/U_L = 1/h_o + R_f,o + R_w + (D_o/D_i) R_f,i + (D_o/D_i)/h_i	h_c = 4753.9, alpha_sc = 1641.6	U_L = 719.94 W/m2.K	HSB Eq. 18.23
Zone 3: mean temperature difference	dT = [(T_sat - T_c') - (T_h,out - T_c,in)] / ln[(T_sat - T_c')/(T_h,out - T_c,in)]	[(61.85 - 26.169) - (50 - 25)] / ln[...]	dT_sub = 30.024 K	HSB Eq. 18.23
Zone 3: area	A_sub = Q_sub / (U_L dT_sub)	1.4931e+05 / (719.94 30.024)	A_sub = 6.9075 m2	HSB Eq. 18.23
Total required area	A_req = A_dsh + A_con + A_sub	0 + 71.757 + 6.9075	A_req = 78.664 m2	HSB Eq. 18.24; Lecture 24
Required tube length	L_req = A_req / (pi D_o N)	78.664 / (pi 0.0254 247)	L_req = 3.9911 m	HEDH 3.4.9B
Length of the condensing zone. Any excess area is shared in proportion, so the zones fill the tube	L_con = (A_con / (pi D_o N)) (L / L_req)	(71.757 / (pi 0.0254 247)) (4 / 3.9911)	L_con = 3.6488 m	Module, as plate.py
Two-phase friction over the condensing zone: Friedel, integrated along the tube on the x(z) found from the heat transfer. D_i = 0.02118 m	dP_F = integral (-dp_F/dz) dz	trapezoid, 40 steps over 3.6488 m	dP_F = 752.55 Pa	HSB Sec. 18.4.1; Eq. 10.49
Acceleration term. Negative: recovery as the vapour decelerates	dP_a = m^2 {[(1 - x)^2/(rho_L (1 - eps)) + x^2/(rho_G eps)]_out - [...]_in}	m = 57.455, Premoli void fraction	dP_a = -513.38 Pa	HSB Eq. 10.27
Gravity term, sin(alpha) = -1 for downflow. Negative: a pressure gain	dP_g = integral g sin(alpha) [rho_G eps + rho_L (1 - eps)] dz	sin(alpha) = -1, Premoli void fraction, 3.6488 m	dP_g = -2455.5 Pa	HSB Eq. 10.26
Net condensing-side pressure change in the tubes	dP_net = dP_F + dP_a + dP_g + dP_dsh	752.55 + (-513.38) + (-2455.5) + 0	dP_net = -2216.3 Pa	HSB Eq. 10.27; HEDH 3.4.7 Eq. (1)
Condensing-side drop compared with the allowable: recoveries not credited, because HEDH 3.4.7 calls omitting them conservative	dP_cond = dP_F + dP_dsh + max(dP_a, 0) + max(dP_g, 0)	752.55 + 0 + 0 + 0	dP_cond = 752.55 Pa	HEDH 3.4.7
Cross-check: required area with Boyko-Kruzhilin as the shear term, same tube count	alpha_sh = alpha_LO [1 + x (rho_L/rho_G - 1)]^(1/2), alpha_LO = 0.021 (lambda_L/D) Re_LO^0.8 Pr_L^0.43	Kosky-Staub A_req = 78.664 m2	A_BK = 80.421 m2	HSB Eqs. 16.54, 16.55
Sensitivity: required area with Chen's F as the shear term, HSB's interim suggestion	alpha_sh = F 0.023 (lambda_L/D) Re_L^0.8 Pr_L^(1/3)	Kosky-Staub A_req = 78.664 m2	A_Chen = 85.924 m2	HSB Eqs. 11.49 to 11.51, Sec. 16.3.2.2
Sensitivity: required area with the suction correction on	tau_0m = tau_0 phi/(exp(phi) - 1)	this run, suction off: A_req = 78.664 m2	A_suction = 77.282 m2	HSB Eq. 16.76

Step	Formula	Numbers	Result	Source
Comparison: dry-zone area on HSB's simplest conservative assumption, no condensation until the vapour reaches T_sat	A = M_v c_pv (T_h,in - T_sat) / (U_s LMTD)	79870 / (203.99 26.057); Bell zones: A_dsh = 0 m2	A_dsh,cons = 15.026 m2	HSB Sec. 18.3.2
Shell wall for pressure, corroded inside radius R_v = D_s/2 + c. S = 121 MPa (HEDH 4.3.6, SA-515-70 carbon steel plate at 100 degC)	t = P R_v / (S E - 0.6 P)	5e+05 (0.3057) / [1.21e+08 (0.85) - 0.6 (5e+05)]	t_shell = 0.0014905 m	HEDH 4.3.6 Eq. (8), quoting ASME UG-27
Shell wall to use: the larger of pressure plus corrosion and the TEMA class R minimum for carbon and low alloy steel	t_use = max(t + c, t_TEMA)	max(0.0014905 + 0.003, 0.0095)	t_shell,use = 0.0095 m	HEDH 4.3.6 Table 2; HEDH 4.2.6-1
Tube wall for internal pressure. Seamless, so E = 1	t = P R / (S - 0.6 P), R = D_i/2	5e+05 (0.01059) / [1.21e+08 - 0.6 (5e+05)]	t_tube = 4.3869e-05 m	HEDH 4.3.6 Eq. (8), quoting ASME UG-31
Tubesheet, fixed: a rough estimate only, because HEDH lists the fixed-tubesheet clauses and does not print the formula	L_ts = 0.1 D_s	0.1 (0.6054)	L_ts = 0.06054 m	HEDH 3.3.5 Eq. (1); HEDH 4.3.2I
Vibration: effective tube mass. Tube bore taken full of condensate, the heaviest case, so the lowest frequency. k read from the chart, +-0.02. E and density of 18Cr/8Ni stainless (S30400), HEDH 5.5.12 at 100 degC	M_e = rho_t (pi/4) D_i^2 + rho_m (pi/4)(D_o^2 - D_i^2) + k rho_s (pi/4) D_o^2	582.9 (pi/4) 0.02118^2 + 7900 (pi/4)(0.0254^2 - 0.02118^2) + 2.9 (994.74) (pi/4) 0.0254^2	M_e = 2.8867 kg/m	HEDH 4.6.2A, Fig. 3; HEDH 5.5.12
Vibration: natural frequency of the longest span, L = 0.48432 m. C_n = 10, the lowest curve of the chart, conservative for any span count, equal spans. No axial load	f_n = (C_n / 2 pi) [E I / (M_e L^4)]^(1/2), I = pi (D_o^4 - D_i^4)/64	(10 / 2 pi) [1.9e+11 (1.0554e-08) / (2.8867 (0.48432)^4)]^0.5	f_n = 178.83 Hz	HEDH 4.6.2 Eq. (1), Fig. 2
Vibration: shell-side velocity, 80% of the coolant flow through the tube gaps of the row at the baffle tip (HEDH 4.6.3A)	u_c = 0.8 M_c / (rho L_bc (chord/L_tp,eff)(L_tp - D_o))	gap area = 0.023105 m2	u_c = 1.064 m/s	HEDH 4.6.3
Vibration: critical velocity for fluid elastic instability. beta = 2, the lowest band of the chart for every layout, because the scan does not show clearly which upper band is which, +-0.1. delta_o = 0.1 for liquids	u_crit = beta f_n D_o [M_e delta_o / (rho_s D_o^2)]^(1/2)	2 (178.83) (0.0254) [2.8867 (0.1) / (994.74 (0.0254)^2)]^0.5	u_crit = 6.0928 m/s	HEDH 4.6.4 Eq. (4), Fig. 4
Vibration: vortex shedding frequency. Sr read from the chart at this pitch ratio and layout, +-0.03	f_vs = Sr u_c / D_o	0.415 (1.064) / 0.0254	f_vs = 17.384 Hz	HEDH 4.6.4 Eq. (2), Fig. 2
Vibration: turbulent buffeting frequency. p_l and p_t from the layout	f_tb = (u_c D_o / (p_l p_t)) [3.05 (1 - D_o/p_t)^2 + 0.28]	(1.064 0.0254 / (0.03175 0.03175)) [3.05 (1 - 0.0254/0.03175)^2 + 0.28]	f_tb = 10.777 Hz	HEDH 4.6.4 Eq. (3)

Convergence

Iteration	Converged	Passes	Tolerance	Residual
Number of tubes	True	12	0.5	0
Step table at the final tube count (a table, not an iteration)	True	40	0	0

Design basis

- Configuration: vertical in-tube downflow, coolant on the baffled shell side (HSB Fig. 17.4). HEDH 3.4.3A: the shell side is baffled and the coolant flows in a single pass countercurrent to the vapour; HSB Figs. 17.4, 18.6 show the same. A cocurrent coolant (HEDH says it gives better subcooling control) is not offered, because HSB's zone equations are written for countercurrent flow.

- One tube pass in every variant: a second pass would turn in-tube downflow into upflow and mix the phases at the turnaround, the reason HSB Sec. 18.2.1 gives against multipass horizontal tube-side condensers; HEDH 3.4.9F calls a single coolant pass the usual arrangement outside vertical tubes. The tube-count correction for several passes exists only as a chart, HEDH Fig. 3.3.5(10).

- Properties are constant: saturated at T_sat for the condensing side, mean temperature for the coolant. The user enters the coolant properties for that temperature; the module cannot verify this.

- Pressure drop does not feed back on T_sat in the iteration; a check reports the size of the effect (HEDH 3.4.9B step 7).

- The tube length entered is the effective length between tubesheets, L_ti = L_ta for a fixed tubesheet (HEDH 3.3.5 Eqs. 3, 5). The tubesheet thickness is reported but not taken off it.

- Clearances: an entered value wins. Blank L_bb takes HEDH Fig. 3.3.5(14) at the bundle type and blank L_tb takes Fig. 3.3.5(12) (TEMA as quoted, drilled holes), because HEDH says to use them without drawings. Blank L_sb takes Eq. (18), the curve through the TEMA values, and not Eq. (19), which adds 1.5 mm that HEDH says is often added for out-of-roundness.

- Mechanical results are preliminary sizing from HEDH Part 4, which quotes ASME Section VIII and TEMA. They do not establish code compliance.

- Condensing coefficient: alpha = (alpha_sh^2 + alpha_grav^2)^(1/2), HSB Eq. 16.56, at each step. Kosky-Staub is the shear term: it is HSB's stated procedure (Sec. 16.3.2, Ex. 16.2) and carries the pressure gradient. Boyko-Kruzhilin needs no shear calculation but HSB says it has 'a much weaker fundamental basis', so it is the recorded cross-check. Chen's form, which HSB suggests as an interim measure, is the low sensitivity case.

- Suction correction (HSB Eq. 16.76) is OFF, the default. HSB calls it usually necessary at larger condensation rates, and Ex. 16.6 raises alpha by 36%, but HSB also argues the correlation's underestimate of pressure gradient offsets an overestimate of film theory, and Fig. 16.9 shows predictions above data. Off gives the lower coefficient, so the larger area. The other setting is run at the final count and reported.

- No entrainment and a thin film, as in HSB Ex. 16.2. Friedel in the printed form for all orientations: HSB and HEDH 2.3.2 both say a slightly different correlation applies to downflow and neither gives it, and HSB uses this form in its own downflow Ex. 16.2. The two books print different constants, 3.24 and (1 - x)^0.24 in HSB Eqs. 10.49, 10.51 against 3.23 and (1 - x)^0.224 in HEDH 2.3.2 Eqs. 46, 48. The code uses HSB's, because its worked examples do and they are the benchmarks.

- Desuperheating by the zones of HSB Fig. 18.6 (Bell 1986): a dry-wall zone until the wall reaches T_sat (Eq. 18.13), then the wet-wall load is added to the condensing zone at T_h = T_sat with the saturated-vapour coefficient, which HSB calls a first approximation supported by experiment (Eqs. 18.17 to 18.19). Eq. 18.10 is written by HSB for vapour outside the tube; here it is applied on the inside area, with U and alpha on the same area, which leaves the algebra unchanged. The Harwell annular-flow model (Whalley et al.) that HSB describes is 'probably far too complicated' per HSB and is not used.

- Subcooling zone as a falling film: HEDH 3.4.3A says the condensate below the condensing zone 'is subcooled as a falling film', and 3.4.6G gives Eqs. 31 to 33, repeated on Lecture 24. HSB Fig. 18.6 draws a flooded tube and gives no coefficient for it; HEDH 3.4.3A does not recommend flooding (poorer subcooling, no venting, harder pressure control, steam hammer).

- Why it iterates: the tube count sets the condensing mass flux, so alpha, U, the required area and the two-phase pressure drop; and through the bundle diameter it sets the shell-side velocity, coefficient and pressure drop. So the area cannot be found before the count (HEDH 3.4.9B: 'assume the number and diameter of tubes and then calculate the required length'). First guess: N = 8, HEDH's rule of thumb for the fewest tubes in a pass, starting low because each update can only raise the count until the limits are met. Update: while any limit is exceeded and no acceptable count is known, N(next) = N x worst ratio, rounded up, where the ratios are L_req/L, (dP/dP_allow)^(1/1.75) from dP ~ m^1.75 (the coolant on the shell side uses the first power, since a larger bundle changes its velocity and its row count together), and velocity/limit. These only need to point the right way, because the bisection below corrects any overshoot. Once an acceptable count exists, bisect between the largest count that failed and the smallest that passed. Stop: those two differ by one, or the update repeats. The returned count always passed; it is the smallest one if feasibility is monotonic in N, which holds for length and tube-side limits but is not proven for the shell-side pressure drop.

Scope limits

- No source works a complete vertical shell-and-tube condenser example: not HSB Ch. 16 to 18, HEDH 3.3, 3.4 or Lectures 9 to 12, 23, 24. The condensing path has only sub-path benchmarks, limits, closure and trends. Eq. 16.56 itself has no worked example.

- Friedel for downflow: HSB and HEDH 2.3.2 name a separate form and neither gives it. The printed form is used, as HSB itself does in its downflow Ex. 16.2.

- Tube count: HEDH Eq. 3.3.5(16) is an estimate, within about 5% for one pass. The design is not re-run on a standard shell: the next shell of HEDH 4.2.5 Table 2a and its count are reported, so round up and rate with a fixed count when a shell is chosen. Table 2a is coded for its five triangular layouts only.

- Non-condensables: HEDH 3.4.9B gives only a first approximation below 5 wt% and says there is no simple method between 5 and 80 wt%, so the module refuses 5 wt% and above. HEDH gives no vent size or vent rate, and neither does HSB, so none is calculated.

- Mechanical: HEDH does not print the fixed-tubesheet formula (0.1 D_s is used), does not define D_L in the tubesheet shear equation (only its criterion is checked) and does not reproduce the ASME chart factor for external pressure on tubes. Expansion joints, flanges, nozzles and supports are not sized.

- Vibration: the HEDH 4.6.5 screen predicts vibration, not damage, and HEDH reports about 75% correct predictions on field cases. HEDH 4.6 has no worked example, so this path has only hand values. The amplitude Eq. 4.6.2(4) is not used: its g_c is listed as 9.81 m/s2 with E in N/m2, which does not close in SI.

- Entry and exit losses on the condensing side: HSB Sec. 18.4.3 and HEDH 3.4.7 say to use the homogeneous model with a multiplier given elsewhere. No nozzle or header size is an input, so they are not calculated.

- Bell-Delaware: J and R factors are the curve fits HEDH gives for computer use (Eqs. 3.3.6(20), (23), (25), (26)), and j_i, f_i are its Table 1 fits. Lecture 12 presents an older chart form of the method (F_n, F_w, F_b, F_L, with alpha values 1.5, 1.35, 5, 4), which is not used because HEDH's equations benchmark end to end. Viscosity ratio at the wall is an input, not iterated with the wall temperature. Laminar window drop: HEDH suggests proration between Re 50 and 200; no proration is done.

- Kern's method is only the cross-check on the coefficient: HSB's friction factor is Fig. 6.6 and Lecture 11's are charts, with no equation, and Lecture 11 says the method performs poorly on pressure drop. HSB limits it to a 25% cut.

- Superheated-vapour properties are taken at saturation for density and viscosity; the vapour specific heat and conductivity are as entered. Pressure drop in the falling-film subcooling zone is not calculated: the bore is open and no source gives a value.

References

- Hewitt, G.F., Shires, G.L. and Bott, T.R., Process Heat Transfer, CRC Press / Begell House, 1994.

- Heat Exchanger Design Handbook, Hemisphere Publishing (Hewitt, ed.).

- Das, S.K., ME6220 Heat Exchanger Design lecture slides, IIT Madras, 2026 (Lectures 1-28).